Model 2: Gradient Boosting (XGBoost) – Maggie

1. Purpose

Model 2 asks whether behavioral characteristics predict 1-year percent weight change beyond demographics and starting BMI, and if so, which behaviors matter most.

The sample is 2,794 U.S. adults who tried to lose weight (NHANES Aug 2021–Aug 2023). The outcome is pct_wt_change, the percent change from weight one year ago to current weight. Negative values mean weight was lost.

XGBoost (extreme gradient boosting) builds many small decision trees in sequence, each correcting the errors of the ones before it, so it can capture non-linear effects and interactions without these predictors being specified in advance.

2. Predictor sets

XGBoost is fit twice:

- **Covariates only (15):** age, sex, race/ethnicity, education, income-to-poverty ratio, marital status, household size, U.S.-born, veteran status, and BMI one year ago.
- **Covariates + behaviors (27):** the 15 covariates plus 12 behaviors: moderate and vigorous activity (min/week), sedentary hours/day, depressive symptoms (PHQ-9), drinks/week, sleep hours, smoking status, and meal-planning and food-shopping roles.

3. Steps

- **Train/test split.** The data are split 80/20 (2,235 training, 559 test). The split is balanced on the outcome so both sets get a similar range of weight change. The test set is used only for the final evaluation.
- **Missing values.** No imputation is needed. XGBoost handles missing values natively by learning, at each split, which branch people with a missing value should go down.
- **Hyperparameter tuning.** A randomized search tries 60 combinations of settings with 5-fold cross-validation on the training set, separately for each predictor set. The search covers the number of trees, learning rate, tree depth, minimum leaf size, row and column subsampling, and regularization. Because the outcome is noisy and the sample is modest, the search favors shallow trees and strong regularization to avoid overfitting. Both final models ended up with trees of depth 3 and small learning rates (about 0.007–0.012).
- **Performance.** Models are scored on the test set with R^2, RMSE and MAE.
- **Do behaviors add value?** This is checked in two ways:
    - A paired bootstrap on the test set (2,000 resamples) gives the change in R^2 and RMSE with 95% confidence intervals.
    - Repeated 5-fold cross-validation (5 repeats, 25 folds) on the training set, with each model's tuned settings held fixed.
- **Which predictors matter?**
    - **Permutation importance:** how much test R^2 drops when one predictor's values are randomly shuffled. A large drop means the model relies on that predictor.

    - **Grouped permutation:** all 12 behaviors are shuffled together (and, separately, all 15 covariates), measuring the total contribution of each block.
    - **SHAP values:** for each person, how much each predictor pushed their prediction up or down. These show the direction and shape of each effect, and SHAP interaction values identify which predictor pairs interact most.
- **Residual check.** Plots of observed vs. predicted values and of residuals.

4. Key results

Predictors	CV R^2 (5×5)	Test R^2	Test RMSE
Covariates	0.127	0.065	8.78
Covariates + behaviors	0.138	0.075	8.74

- **Behaviors add a small, fairly consistent gain.** In cross-validation, R^2 rose by about 0.011 when behaviors were added, an improvement in 22 of 25 folds. On the test set the gain was 0.010, but the 95% CI includes zero (−0.015 to 0.033), so it is not statistically certain.
- **Behaviors as a block:** shuffling all 12 behaviors together lowered test R^2 by 0.024 (95% range −0.002 to 0.047). Shuffling the covariates lowered it by 0.246, mostly because of starting BMI.
- **Starting BMI dominates**, by a wide margin (mean |SHAP| 2.55, vs. 0.47 for the next predictor, age).
- **Most important behaviors** (by mean |SHAP|): moderate activity, PHQ-9, sedentary hours, sleep, and vigorous activity, followed by drinks/week, current smoking and main meal planner. Meal-planning and food-shopping sharing contributed very little.
- **Shape of the behavioral effects** (from SHAP dependence plots):
    - **Moderate activity:** more minutes go with more weight loss, but the benefit levels off after about 300 minutes/week.
    - **Depressive symptoms (PHQ-9):** higher scores go with less weight loss.
    - **Sedentary time:** a step pattern. People sitting more than about 5 hours/day had less weight loss than those sitting less.
    - **Sleep:** sleeping less than about 6–6.5 hours/night went with less weight loss. Beyond about 7 hours there was little further difference.
- **Overall explanatory power is modest** (R^2 about 0.07–0.14). That's expected for self-reported 1-year weight change, which is noisy.
- **Interactions:** the strongest interactions involved starting BMI, with income, moderate activity, age and PHQ-9. This suggests that some behavioral effects differ depending on starting BMI.

5. Limitations

- Importance measures describe what the model relies on, not causal effects. The data are cross-sectional.
- Behaviors were measured at the exam, at the end of the weight-change period, so the direction of influence is unclear.

- Correlated predictors (e.g., meal planning and food shopping) share credit, so individual importances can understate them. The grouped importance is more reliable for the behavior block as a whole.
- The models are unweighted, so results describe this sample rather than the U.S. population. NHANES survey weights can be applied by setting USE_SURVEY_WEIGHTS = True in the notebook.